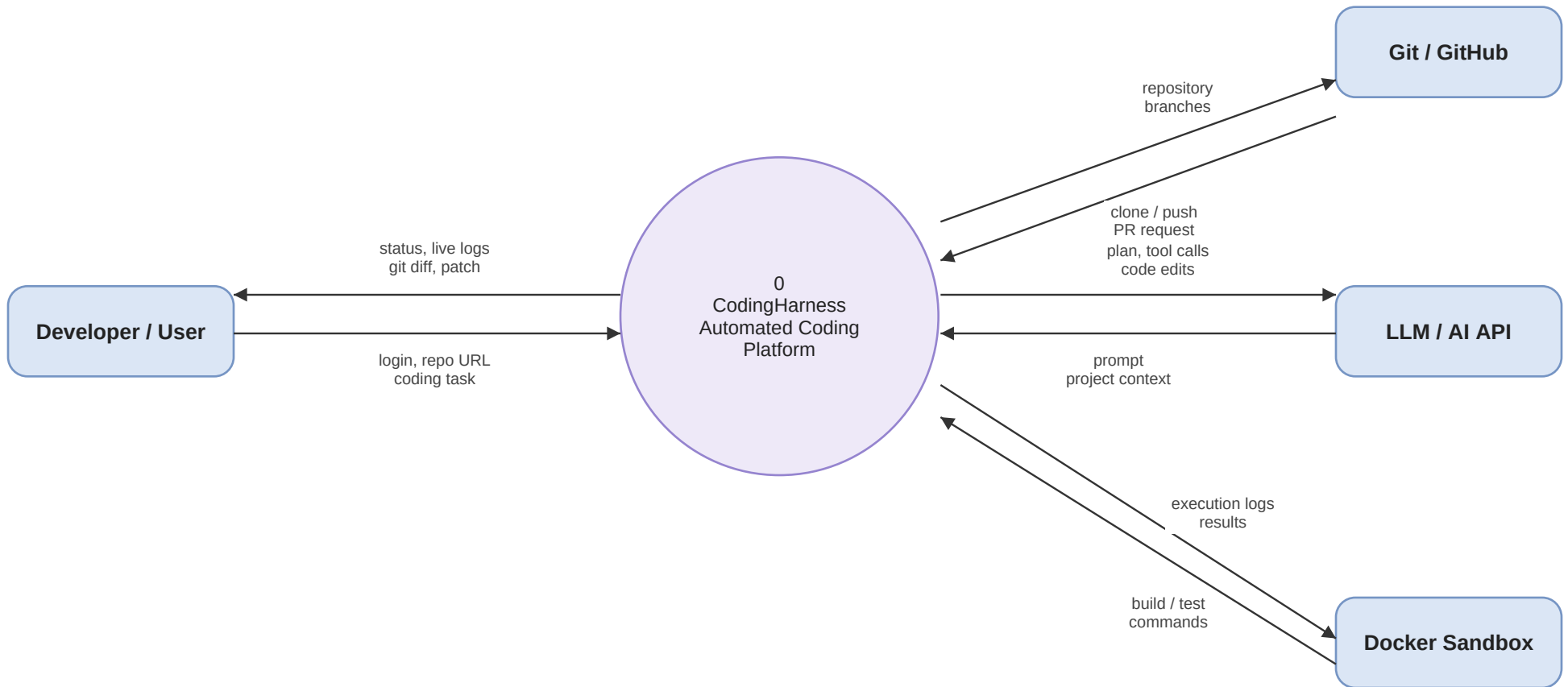
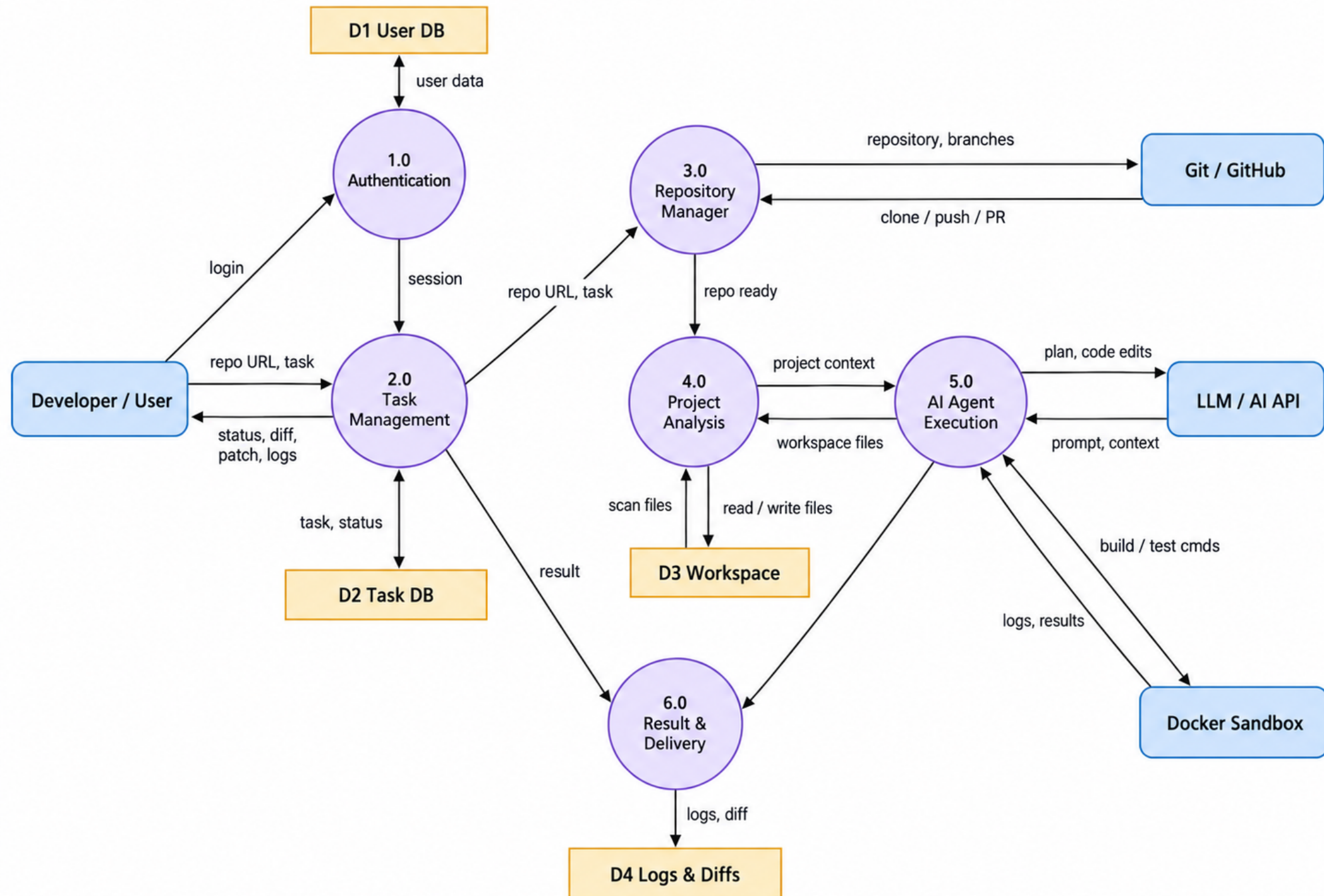


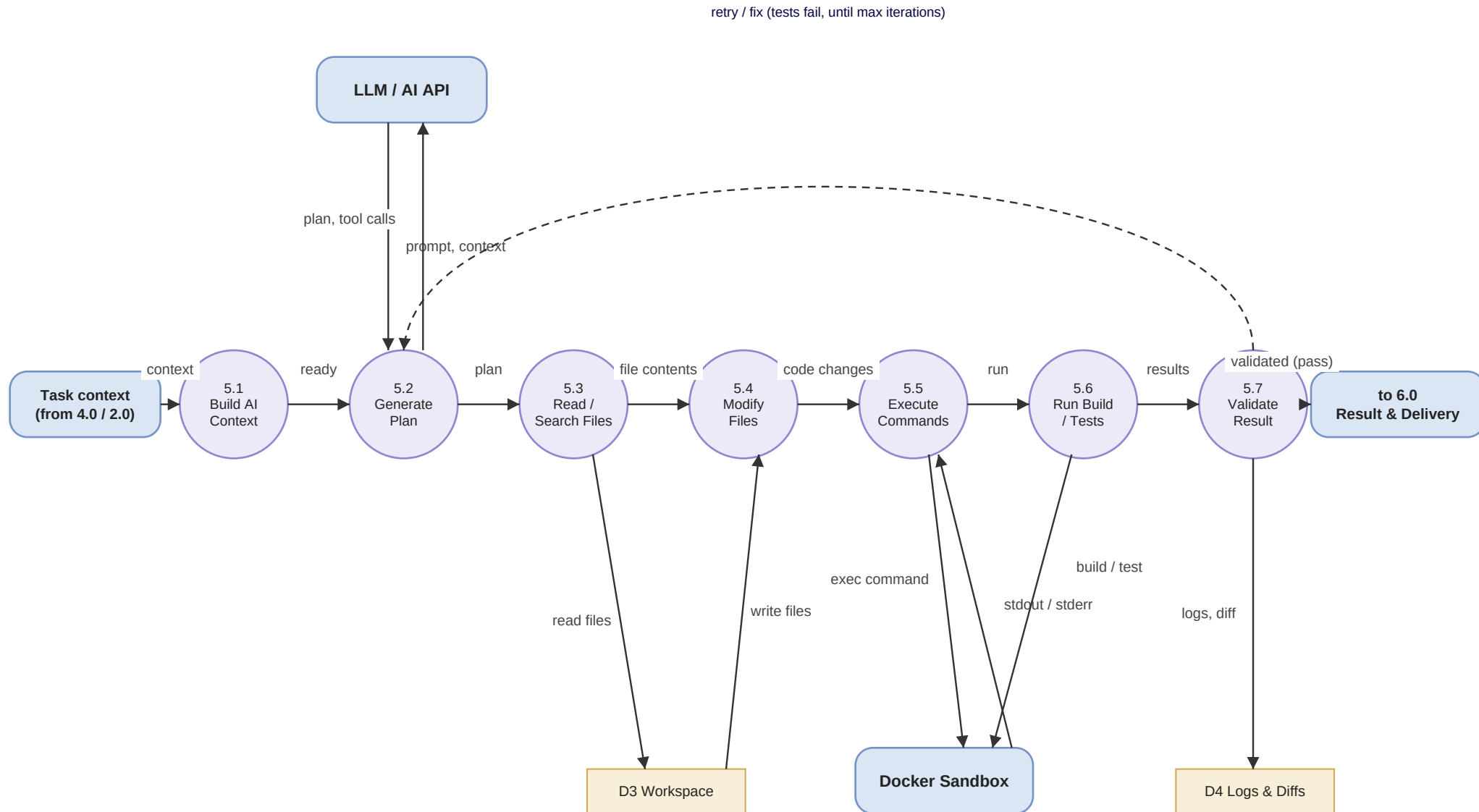
# DFD Level 0 - Context Diagram



# DFD Level 1 - CodingHarness System Processes



# DFD Level 2 - AI Agent Task Execution (Process 5.0)



# DFD - Notes

Level 0 (Context) shows CodingHarness as a single process interacting with four external entities:

Developer / User, Git / GitHub, LLM / AI API, and Docker Sandbox.

Level 1 decomposes the platform into six processes - Authentication, Task Management, Repository Manager, Project Analysis, AI Agent Execution, and Result & Delivery - over four data stores: D1 User DB, D2 Task DB, D3 Workspace, and D4 Logs & Diffs.

Level 2 expands the AI Agent Execution process (5.0) into its sub-processes: build AI context, generate plan (LLM), read / search and modify files (D3), execute commands and run build / tests (Docker Sandbox), validate result, and the retry / fix loop that repeats on test failure until success or the maximum iteration limit is reached.